



Innopolis University

Homework 1: Customer Interview

Business Analyst Role

Application: hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Submission: June 16, 2026

1. Statement of Task

Objective: To conduct a user research study in the role of a business analyst for the **hh.ru** platform by performing in-depth customer interviews with the second team acting as users of the selected application.

The selected application was agreed by both teams before the interview. The respondent team tested the main functions of hh.ru before answering the interview questions, including vacancy search, resume review, cover-letter preparation, and application-status review.

Key steps of the assignment:

1. Prepare a list of 10 guesses for improving the selected application.
2. Select 3 guesses and formulate 3 hypotheses using the structured format from the lecture.
3. Describe the target group of respondents and their common characteristics.
4. Develop a list of open-ended interview questions aimed at validating or rejecting the formulated hypotheses.
5. Conduct an interview with representatives of another team and record their responses in a question-and-answer format.
6. Analyze the results and draw conclusions about whether each hypothesis was confirmed, partially confirmed, or refuted.



2. Ten Guesses for App Improvement

Before formulating the guesses, the current functionality of hh.ru was reviewed to avoid suggesting features that already exist on the platform and to focus on real limitations in the current user experience.

1. **Deep personalization of AI-generated cover letters.** hh.ru already includes a “Generate Letter” button, but the resulting text may sound generic and barely reflect the specific vacancy requirements. The improvement is to generate letters based on the full vacancy text, company profile, and candidate background, with several tone and structure options.
2. **Merging duplicate job listings from the same employer.** Large companies may post the same vacancy separately for different cities, which floods search results with similar cards. The improvement is to consolidate such listings and show available locations inside one vacancy card.
3. **Practical skill verification.** Current skill confirmations are mostly test-based. The improvement is to verify applied skills through practical tasks, code repositories, case studies, or project reviews.
4. **Filtering out inactive or ghost jobs.** The improvement is to show whether the employer actively reviews applications and responds to candidates, so users can avoid vacancies that remain open for months without real hiring activity.
5. **AI-powered mock interview trainer.** The improvement is to let candidates practice interviews with an AI assistant and receive feedback on answer structure, clarity, and soft skills.
6. **Personal market value tracker.** The improvement is to monitor the candidate’s estimated market value over time and provide recommendations for skills or courses that may increase it.
7. **Advanced relocation filters.** The improvement is to add filters for destination country, visa support, relocation package, and legal employment conditions.
8. **Full anonymous search mode.** The improvement is to hide the user’s resume not only from selected employers, but also from subsidiaries and affiliated entities within the same corporate group.
9. **Portfolio integration into the candidate profile.** The improvement is to let candidates connect GitHub, Behance, Figma, or similar platforms directly to their hh.ru resume and display projects as a structured portfolio block.
10. **Transparent rejection reasons.** The improvement is to show candidates structured rejection reasons, such as missing skills, experience mismatch, salary mismatch, or failed test task, instead of only a generic rejection status.

3. Selected Hypotheses

Guesses 1, 9, and 10 were selected for validation because they are directly connected to the problems of students and junior candidates searching for internships or entry-level jobs. The hypotheses are formulated according to the lecture template:

I believe (who) is facing a problem of (what) (how much/when) because (why).

3.1 Hypothesis 1: AI Cover Letter Personalization

Statement: I believe second-year undergraduate students aged 19–20 with no commercial work experience, actively applying for internships and Junior IT positions on hh.ru, are facing a problem of writing cover letters that are specific enough for each vacancy when they apply to several positions per day because the current letter generation tool produces generic text that still requires manual rewriting and does not deeply use the vacancy requirements or the candidate’s academic background.

3.2 Hypothesis 2: Portfolio Integration

Statement: I believe second-year undergraduate students aged 19–20 with no commercial work experience are facing a problem of demonstrating practical skills in their hh.ru resume when applying for internships or Junior IT positions because they rely on projects instead of commercial experience, but the standard resume format does not provide a convenient native way to connect and display GitHub, Behance, Figma, or other portfolio platforms.

3.3 Hypothesis 3: Transparent Rejection Reasons

Statement: I believe second-year undergraduate students aged 19–20 applying for their first internships or Junior IT positions are facing a problem of understanding how to improve their applications after repeated rejections because hh.ru usually shows only a generic rejection status and does not provide actionable employer feedback, such as missing skills, insufficient experience, salary mismatch, or weak test-task performance.

4. Target Respondent Description

The respondents are members of another student team from Innopolis University. The application hh.ru was agreed by both teams as the product for the interview. The respondents tested the main functions of hh.ru for at least two days before the interview and answered the questions from the perspective of job seekers.

Common characteristics of the respondent group:

- **Age and status:** Undergraduate students aged approximately 19–20 years old.
- **Education:** Students in a technology-oriented university environment.
- **Digital behavior:** Comfortable with online platforms, digital forms, AI tools, and job-search services.
- **Job search context:** Interested in internships, Junior IT positions, or first professional opportunities in the technology industry.
- **Experience level:** Limited or no commercial work experience, so resumes, cover letters, and portfolios are especially important for them.
- **Main need:** Fast, clear, and useful feedback from the platform because they are still learning how to present themselves to employers.

5. Interview Questions

The questions below were designed as open-ended questions. They avoid directly suggesting the desired answer and focus on the respondent's real experience.

5.1 Block 1: Testing Hypothesis 1 – Cover Letter Personalization

1. Tell us about the last time you applied or considered applying for a vacancy on hh.ru. Did you need a cover letter?
2. How do you usually decide what to write in a cover letter?
3. What part of writing a cover letter is the most difficult for you?
4. Have you used automatic text generation for cover letters? What was useful or not useful about it?
5. What makes a cover letter feel suitable for a specific vacancy in your opinion?

5.2 Block 2: Testing Hypothesis 2 – Portfolio Integration

1. How do you usually show your projects, coursework, GitHub repositories, designs, or other practical work to employers?
2. What information in your resume do you think best proves your skills?
3. Have you ever had difficulties adding project links or portfolio materials to a resume?
4. How do you think recruiters usually evaluate student projects or pet projects?
5. What would make it easier for you to present your projects inside a job-search platform?

5.3 Block 3: Testing Hypothesis 3 – Transparent Rejection Reasons

1. Tell us about a situation when your application was rejected or ignored. What information did you receive?
2. How do you usually understand what went wrong after a rejection?
3. What do you change in your resume or cover letter after receiving a rejection?
4. What kind of feedback would be useful for you after an unsuccessful application?
5. How would detailed rejection reasons affect your motivation to continue applying?

6. Interview Execution

The interview was conducted with three respondents from another student team. The discussion lasted more than 10 minutes. The interview notes were shared within the team for further analysis. The most relevant answers are recorded below in a question-and-answer format.

6.1 Respondent 1

Q: Tell us about the last time you applied or considered applying for a vacancy on hh.ru. Did you need a cover letter?

A: I looked at several internship vacancies. Some of them had a cover-letter field, and I felt that I should write something, but I was not sure what exactly to include.

Q: What part of writing a cover letter is the most difficult for you?

A: The hardest part is making it sound specific. I can write a general text about myself, but it is difficult to connect it to the vacancy requirements.

Q: How do you usually show your projects to employers?

A: I usually paste GitHub links into the resume or into the “About me” section. It works, but it does not look very structured.

Q: What kind of feedback would be useful after rejection?

A: It would be useful to know whether the problem is lack of experience, missing skills, or something in the resume. Otherwise I do not understand what exactly to improve.

6.2 Respondent 2

Q: Have you used automatic text generation for cover letters? What was useful or not useful about it?

A: I tried using generated text, but it sounded too formal and generic. It could be used as a draft, but I would still rewrite it before sending.

Q: What information in your resume best proves your skills?

A: For me, projects are more important than work experience because I do not have real commercial experience yet.

Q: Have you ever had difficulties adding project links or portfolio materials to a resume?

A: Yes. It is possible to add links manually, but they are not presented as a separate portfolio. A separate project block would be more convenient.

Q: How do you usually understand what went wrong after a rejection?

A: I mostly guess. Sometimes I think the vacancy required more experience, but I cannot know for sure.

6.3 Respondent 3

Q: How do you usually decide what to write in a cover letter?

A: I look at the vacancy requirements and try to mention the skills that match them, but it takes time. If I apply to several vacancies, I do not want to write everything from zero each time.

Q: What would make it easier for you to present your projects inside a job-search platform?

A: I would like to connect GitHub directly, so that the platform shows repositories, descriptions, and technologies automatically.

Q: What do you change in your resume or cover letter after receiving a rejection?

A: Usually nothing specific, because I do not know the reason. I may only change wording randomly.

Q: How would detailed rejection reasons affect your motivation to continue applying?

A: It would make the process less frustrating. Even if the answer is negative, at least I would understand what to work on.

7. Analysis of Hypotheses

Hypothesis	Evidence from interviews	Result
Hypothesis 1: AI cover letter personalization	Respondents reported that cover letters are difficult to adapt to specific vacancies. Generated text was perceived as too generic and requiring manual rewriting.	Confirmed
Hypothesis 2: Portfolio integration	Respondents stated that projects are important because they have limited work experience. They currently add portfolio links manually, which looks unstructured and inconvenient.	Confirmed
Hypothesis 3: Transparent rejection reasons	Respondents said that after rejection they usually receive no useful explanation and have to guess what to improve. They expressed interest in structured feedback categories.	Confirmed

7.1 Hypothesis 1 Conclusion

Hypothesis 1 was confirmed. Students do face difficulties when writing cover letters for specific vacancies. The key problem is not only the time spent on writing, but also the lack of confidence that the letter matches the vacancy. A more personalized AI generator could be valuable if it uses the vacancy requirements and the candidate's background instead of producing a template-like text.

7.2 Hypothesis 2 Conclusion

Hypothesis 2 was confirmed. For students without commercial experience, projects and coursework are a major way to prove practical skills. Manual links in the resume are not enough because they are poorly structured and may be overlooked. Native portfolio integration would make the resume more informative and easier for recruiters to evaluate.

7.3 Hypothesis 3 Conclusion

Hypothesis 3 was confirmed. Generic rejection statuses do not help students improve their applications. Respondents want at least basic categories of rejection reasons, such as missing skills, insufficient experience, weak cover letter, or salary mismatch. This feature could reduce frustration and make the job-search process more educational for junior candidates.

8. Final Conclusions

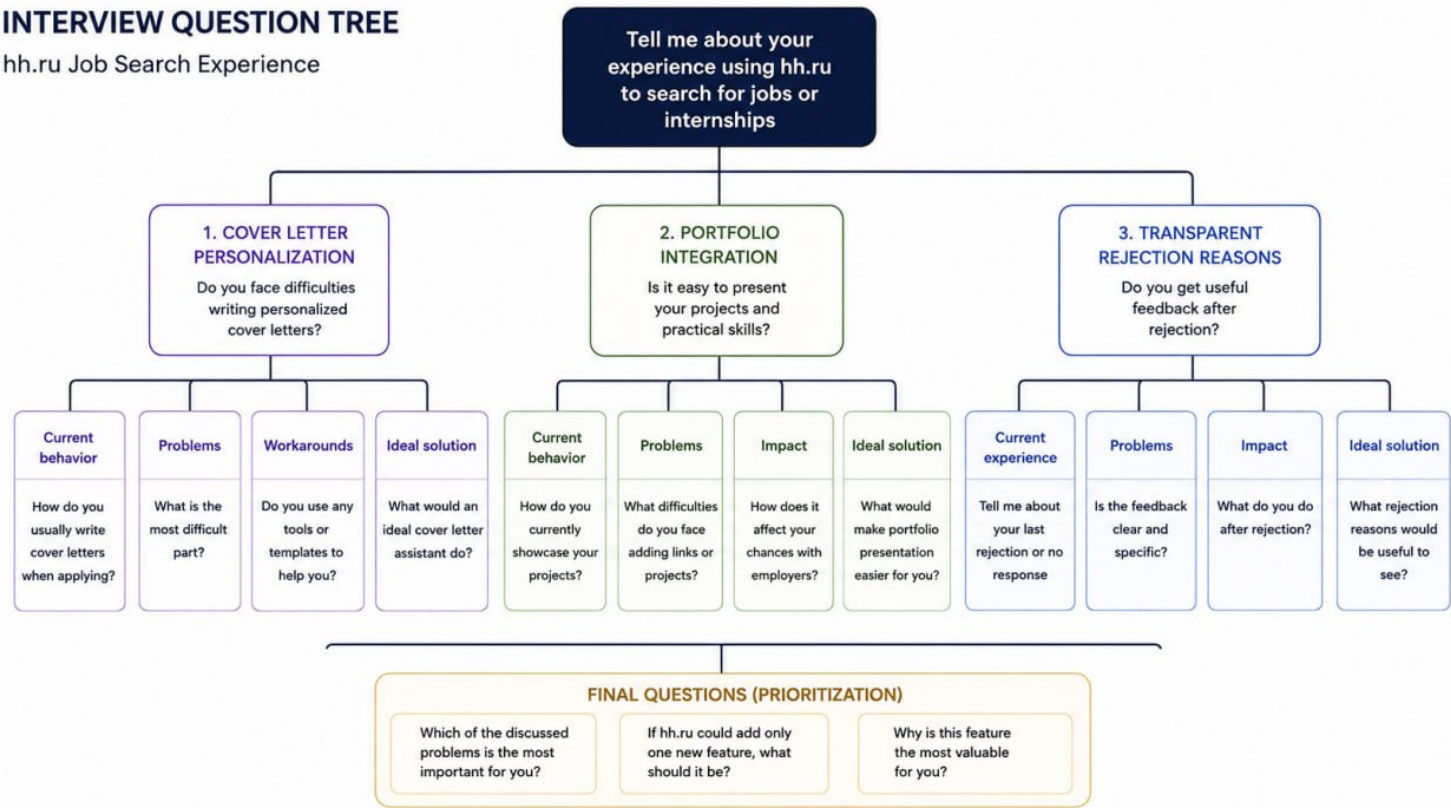
The interviews showed that the selected target group has several important problems when using hh.ru for internship and junior job search.

1. Students need better support when writing cover letters because they struggle to make them specific to each vacancy.
2. Students need a more structured way to present projects because they often lack commercial experience and rely on portfolios to prove skills.
3. Students need more transparent rejection feedback because generic rejection statuses do not help them understand how to improve.

The most promising improvement is **portfolio integration**, because all respondents emphasized that projects are the main proof of skills for students without commercial experience. The second most promising improvement is **transparent rejection reasons**, because it would help candidates learn from unsuccessful applications. The AI cover-letter improvement is also useful, but it should be implemented carefully to avoid producing generic template text.

INTERVIEW QUESTION TREE

hh.ru Job Search Experience





Innopolis University

Homework 2: Use Case Analysis

Business Analyst Role

Activity: Searching and Applying for an Internship or Junior IT Position Using hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Submission: June 23, 2026

1. Statement of Task

Objective: To analyze a daily-life activity from the business analyst perspective, decompose it into at least three dependent use cases, draw a UML use case diagram, and describe each use case using the template from Lecture 6.

Selected activity: Searching and applying for an internship or Junior IT position using **hh.ru**. This activity continues the work from Homework 1, where hh.ru was selected and analyzed through a customer interview.

Key steps of the assignment:

1. Select an activity from daily life and define the system boundary.
2. Separate the activity into at least three dependent use cases.
3. Draw a UML use case diagram with actors, use cases, system boundary, and relationships.
4. Describe each use case using the lecture template, including actors, trigger, preconditions, postconditions, normal flow, alternative flow, exceptions, priority, frequency, assumptions, and other information.
5. Prepare one team homework file in PDF format.

Author role vs. modeled actor: Team 9 works in the **Business Analyst** role. The modeled user is the **Job Seeker**, because the activity describes the candidate-side journey on hh.ru.

2. Activity and System Boundary

System under analysis: hh.ru as a job-search platform.

Primary actor: Job Seeker.

Secondary actor: Employer.

System boundary: The system includes vacancy search, vacancy page review, application submission, application status tracking, and response display inside hh.ru. Employer-side hiring decisions are external to the job seeker journey, but employer messages and status changes are visible through the platform.

Use case decomposition:

- **UC-1 Find a Vacancy** - the Job Seeker searches for and selects a suitable vacancy.
- **UC-2 Submit a Tailored Application** - the Job Seeker applies to the selected vacancy with a resume, cover letter, and portfolio links if needed.
- **UC-3 Review Application Status** - the Job Seeker checks the current application status and any available employer update.

Dependency logic:

1. UC-2 depends on UC-1, because an application can be submitted only after a vacancy has been selected.
2. UC-3 depends on UC-2, because an application status can be reviewed only after an application has been submitted.

3. UML Use Case Diagram

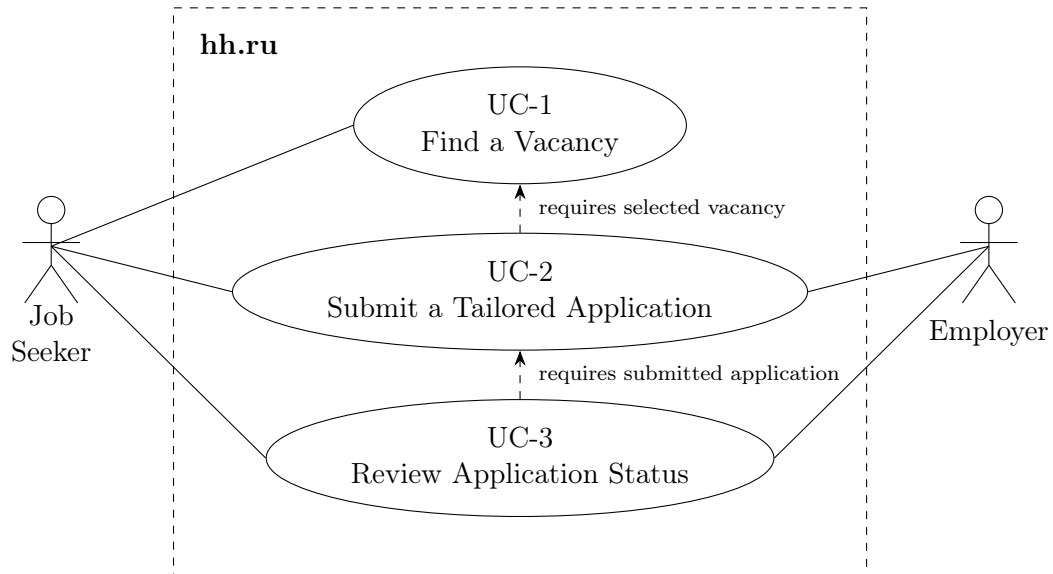
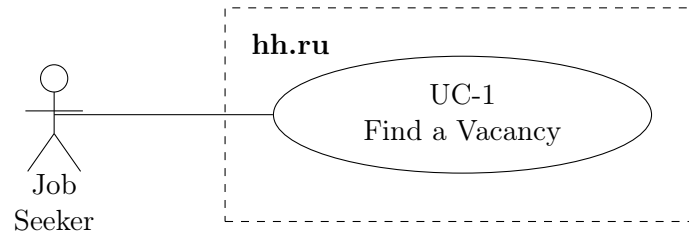


Diagram note: Internal hh.ru services, such as search, recommendations, notification delivery, and cover-letter generation, are not shown as actors because they are parts of the system boundary. The diagram shows only external actors and the dependent user goals.

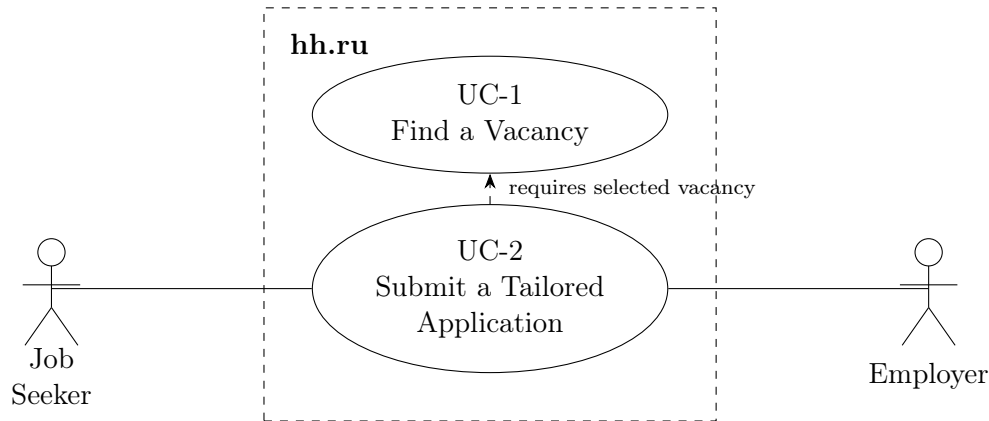
4. Use Case 1: Find a Vacancy



UML note: The diagram shows the external actor and the user goal inside the hh.ru system boundary. Search and recommendation mechanisms are internal system functions, therefore they are not modeled as actors.

Field	Description
ID and Name	UC-1 Find a Vacancy
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	None
Description	The Job Seeker searches for internship or entry-level Junior IT vacancies on hh.ru, applies search filters, reviews vacancy details, and selects one suitable vacancy for possible application.
Trigger	The Job Seeker decides to look for an internship or Junior IT position.
Preconditions	PRE-1. hh.ru is available. PRE-2. The Job Seeker has internet access. PRE-3. The vacancy search database is accessible.
Postconditions	POST-1. A suitable vacancy is selected. POST-2. The selected vacancy details are displayed and reviewed. POST-3. The selected vacancy is available for possible application at the moment of review.
Normal Flow	1.0 Find a Vacancy - The Job Seeker opens hh.ru. - The Job Seeker enters keywords related to the desired position. - The system displays matching vacancies and available search filters. - The Job Seeker applies filters such as city, specialization, experience level, employment type, and salary. - The system updates the list of matching vacancies. - The Job Seeker opens a vacancy card. - The Job Seeker reviews title, company, responsibilities, requirements, salary, location, and employment conditions. - The Job Seeker decides that the vacancy is suitable for application.
Alternative Flow	1.1 Browse Recommended Vacancies - The Job Seeker is logged into hh.ru and opens the recommendation section. - The system displays recommended vacancies based on the profile and previous activity. - The Job Seeker opens a recommended vacancy card. - The flow returns to Normal Flow step 7.
Exceptions	1.0.E1 No Suitable Vacancies Found - The system returns no relevant vacancies. - The system prompts the Job Seeker to change keywords or filters. - The Job Seeker updates search parameters. - The flow returns to Normal Flow step 5. 1.0.E2 Vacancy Page Unavailable - The Job Seeker opens a vacancy that has been deleted or archived. - The system displays that the vacancy is unavailable. - The Job Seeker returns to search results.
Priority	High
Frequency of Use	Multiple times per day during active job search.
Other Information	Vacancy search and viewing can be performed without submitting an application. Login is required for personalized recommendations, favorites, and later application submission.
Assumptions	The vacancy database is updated regularly, and vacancy descriptions contain enough information for the candidate to make a decision.

5. Use Case 2: Submit a Tailored Application



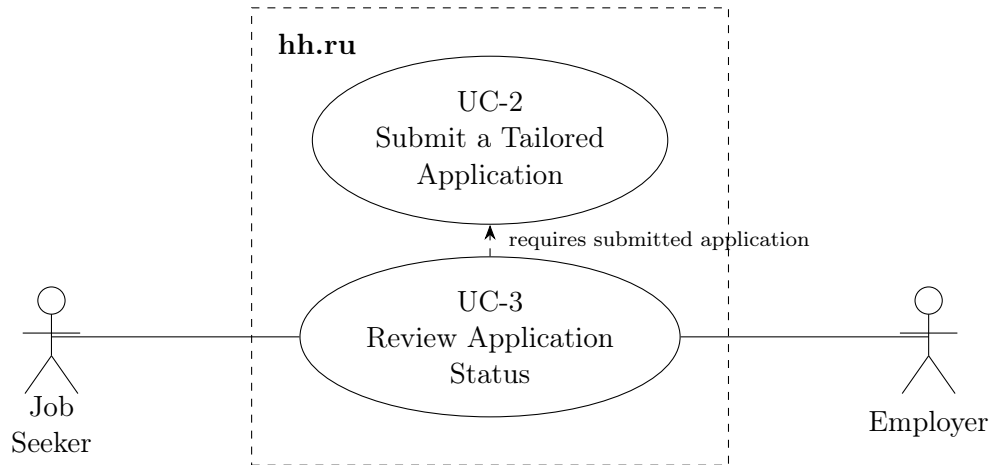
UML note: UC-2 is shown together with UC-1 because the application can be submitted only after a vacancy has been selected. Cover-letter generation is treated as an internal hh.ru feature, not as a separate external actor.

Field	Description
ID and Name	UC-2 Submit a Tailored Application
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	Employer
Description	After selecting a suitable vacancy in UC-1, the Job Seeker submits an application by choosing a resume, preparing or editing a cover letter, adding portfolio or project links if needed, and sending the application to the Employer through hh.ru.
Trigger	The Job Seeker clicks the Apply button on the selected vacancy page.
Preconditions	PRE-1. The selected vacancy is open for applications. PRE-2. The Job Seeker is logged into hh.ru. PRE-3. The Job Seeker has at least one resume in the account. PRE-4. Mandatory profile and resume fields are completed.
Postconditions	POST-1. The application record is created in hh.ru. POST-2. The Employer can see the application in the employer-side interface. POST-3. The application status is updated to sent or another equivalent system status. POST-4. The submitted application is visible to the Job Seeker in the responses section.
Normal Flow	2.0 Submit a Tailored Application 1. The Job Seeker clicks the Apply button on the selected vacancy page. 2. The system prompts the Job Seeker to choose a resume. 3. The Job Seeker selects the most relevant resume. 4. The system opens the application form and cover-letter field. 5. The Job Seeker writes a cover letter or edits the generated draft according to the vacancy requirements. 6. The Job Seeker manually adds portfolio or project links if they are important for the vacancy. 7. The Job Seeker reviews the application content. 8. The Job Seeker submits the application. 9. The system validates the required data. 10. The system sends the application to the Employer and displays a successful submission message.
Alternative Flow	2.1 Use Generated Cover Letter Draft 1. The Job Seeker chooses the built-in cover-letter generation option. 2. The system creates a draft letter. 3. The Job Seeker edits the draft to make it more specific to the vacancy. 4. The flow returns to Normal Flow step 6. 2.2 Add Portfolio Links Manually 1. The Job Seeker pastes GitHub, Behance, Figma, or other project links into the resume or cover-letter field. 2. The Job Seeker adds short explanations for selected projects. 3. The flow returns to Normal Flow step 7.



Field	Description
Exceptions	2.0.E1 Vacancy Closed Before Submission 1. The Employer closes or archives the vacancy before the final submission step. 2. The system informs the Job Seeker that the vacancy no longer accepts applications. 3. The application is not sent. 4. The Job Seeker returns to UC-1 Find a Vacancy. 2.0.E2 Incomplete Resume Data 1. The system detects missing mandatory fields in the selected resume. 2. The system prompts the Job Seeker to complete the missing data. 3. After correction, the flow returns to Normal Flow step 2.
Priority	High
Frequency of Use	Multiple times per day during active job search.
Other Information	The built-in letter generator is optional. Portfolio auto-import is treated as a potential improvement identified in Homework 1, so in the current use case portfolio links are added manually.
Assumptions	The selected vacancy is active, the Job Seeker has valid application materials, and the platform can deliver the application to the Employer.

6. Use Case 3: Review Application Status



UML note: UC-3 is shown together with UC-2 because an application status can be reviewed only after an application has been submitted. Notification delivery is internal to hh.ru and is not modeled as an actor.

Field	Description
ID and Name	UC-3 Review Application Status
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	Employer
Description	After submitting an application in UC-2, the Job Seeker checks the current status of the application and reviews any available employer update.
Trigger	The Job Seeker receives a notification about a status change, receives an employer message, or decides to check the application status manually.
Preconditions	PRE-1. A submitted application exists in the system. PRE-2. The application is visible to the Job Seeker in the responses section. PRE-3. Status tracking is available.
Postconditions	POST-1. The current application status is displayed. POST-2. Any available employer message or status update is displayed. POST-3. The application remains stored in the Job Seeker's responses section.
Normal Flow	3.0 Review Application Status 1. The Job Seeker receives a notification or opens hh.ru to check the application status. 2. The Job Seeker opens the responses section. 3. The system displays the list of submitted applications. 4. The Job Seeker selects the relevant application. 5. The system displays the current status and any available employer message. 6. The Job Seeker reviews the status and response information. 7. The Job Seeker decides on the next action outside the system.
Alternative Flow	3.1 Interview Invitation 1. The system shows that the application status is invitation. 2. The system displays interview details, contacts, or further instructions if the Employer provided them. 3. The Job Seeker reviews the information and prepares for the next hiring stage. 3.2 No Status Change Yet 1. The system shows that the application is still waiting for review. 2. The Job Seeker closes the response page and may continue searching for other vacancies. 3.3 Generic Rejection Comment 1. The system shows that the application status is rejected. 2. The system displays the employer's rejection comment. 3. The comment is generic and does not explain which skills or experience were missing. 4. The Job Seeker reviews the available comment and may revise the resume or continue searching.
Exceptions	3.0.E1 Message Cannot Be Loaded 1. The system fails to load the employer message or response details. 2. The system displays an error. 3. The Job Seeker refreshes the page or returns later.
Priority	High
Frequency of Use	Daily during active job search.
Other Information	A rejection usually includes an employer comment, but the comment can be generic and not actionable. More specific rejection reasons were identified as a possible improvement in Homework 1.



Field	Description
Assumptions	Employers update application statuses within a reasonable time, and hh.ru displays all response information that is available to the candidate.

7. Conclusion

The selected daily-life activity was decomposed into three dependent use cases:

- UC-1 Find a Vacancy;
- UC-2 Submit a Tailored Application;
- UC-3 Review Application Status.

The dependency chain is clear: the Job Seeker first selects a vacancy, then submits an application, and only after that reviews the application status and any employer update. The UML diagram shows the system boundary, external actors, use cases, associations, and dependencies. Each use case is described using the lecture template and includes normal flow, alternative flow, exceptions, preconditions, postconditions, and assumptions.

This structure keeps the analysis focused on the job seeker journey and avoids mixing internal hh.ru services with external actors.



Innopolis University

Homework 3: Business Rules for Use Cases

Business Analyst Role

Process: Searching and Applying for an Internship or Junior IT
Position Using hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Enrolment: June 23, 2026

1. Statement of Task

Objective: To continue the previous homework on use cases and identify business rules that influence the selected process on **hh.ru**.

Selected process: Searching and applying for an internship or Junior IT position using **hh.ru**.

Key steps of the assignment:

1. Reuse the use cases from the previous homework as a basis for the current analysis.
2. Identify relevant business rules for the selected use cases.
3. Specify business rules of different types: facts, constraints, action enablers, inferences, and computations.
4. Document the business rules in a separate table with ID, rule definition, type of rule, and static or dynamic status.
5. Add the IDs of the relevant business rules to the *Business Rules* field of each updated use case table.
6. Prepare one team homework file in PDF format.

Source use cases from Homework 2:

- **UC-1 Find a Vacancy;**
- **UC-2 Submit a Tailored Application;**
- **UC-3 Review Application Status.**

2. Business Rules

The table below documents the business rules that influence vacancy search, application submission, and response review on hh.ru. The rules are grouped by ID prefixes:

- **VAC** - rules related to vacancy search and vacancy selection;
- **APP** - rules related to application submission;
- **RESP** - rules related to employer responses and application status review.

ID	Rule Definition	Type of Rule	Static or Dynamic
VAC-1	Every vacancy has a unique vacancy ID.	Fact	Static
VAC-2	Every vacancy is associated with one employer profile.	Fact	Static
VAC-3	Search parameters include keywords and optional filters, such as city, specialization, experience level, employment type, salary, and work format.	Fact	Dynamic
VAC-4	The system may show only vacancies that are available for viewing and match the active search parameters.	Constraint	Static
VAC-5	If the Job Seeker changes keywords or filters, then the system must update the list of matching vacancies.	Action Enabler	Static
VAC-6	If a vacancy is deleted, archived, or closed by the Employer, then the vacancy is considered unavailable for application.	Inference	Static
VAC-7	The number of matching vacancies is calculated as the count of available vacancies that match the active search parameters.	Computation	Static
APP-1	Every application is linked to one Job Seeker, one selected vacancy, and one resume.	Fact	Static
APP-2	The Job Seeker must be logged into hh.ru to submit an application.	Constraint	Static
APP-3	An application may be submitted only for a vacancy that is open for applications.	Constraint	Static
APP-4	Mandatory profile and resume fields must be completed before the application can be submitted.	Constraint	Dynamic
APP-5	If the Job Seeker requests a generated cover-letter draft, then the system must create a draft based on the available resume and vacancy information.	Action Enabler	Dynamic
APP-6	If the required application data is valid and the Job Seeker confirms submission, then the system must send the application to the Employer and create an application record.	Action Enabler	Static
APP-7	If all mandatory resume fields are completed and the selected vacancy is open, then the application is ready for submission.	Inference	Static
RESP-1	Every submitted application has a current status visible to the Job Seeker.	Fact	Static
RESP-2	Application status may only be one of the predefined values, such as sent, viewed, invitation, rejected, archived, or waiting for review.	Constraint	Dynamic
RESP-3	If the Employer sends a message or changes the application status, then the system must display the update to the Job Seeker.	Action Enabler	Static
RESP-4	Every rejected application has an employer rejection comment visible to the Job Seeker.	Fact	Static
RESP-5	If the rejection comment does not mention specific missing skills, experience, or application problems, then the rejection reason is considered generic.	Inference	Dynamic



ID	Rule Definition	Type of Rule	Static or Dynamic
RESP-6	If the Employer sends an interview invitation, then the system must display interview details and further instructions if they were provided by the Employer.	Action Enabler	Static



3. Updated Use Cases

The following use case tables reuse the Homework 2 template and add the **Business Rules** field. Each field contains the IDs of the business rules that are relevant to the corresponding use case.

4. Use Case 1: Find a Vacancy

Field	Description
ID and Name	UC-1 Find a Vacancy
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	None
Description	The Job Seeker searches for internship or entry-level Junior IT vacancies on hh.ru, applies search filters, reviews vacancy details, and selects one suitable vacancy for possible application.
Trigger	The Job Seeker decides to look for an internship or Junior IT position.
Preconditions	PRE-1. hh.ru is available. PRE-2. The Job Seeker has internet access. PRE-3. The vacancy search database is accessible.
Postconditions	POST-1. A suitable vacancy is selected. POST-2. The selected vacancy details are displayed and reviewed. POST-3. The selected vacancy is available for possible application at the moment of review.
Normal Flow	1.0 Find a Vacancy 1. The Job Seeker opens hh.ru. 2. The Job Seeker enters keywords related to the desired position. 3. The system displays matching vacancies and available search filters. 4. The Job Seeker applies filters such as city, specialization, experience level, employment type, and salary. 5. The system updates the list of matching vacancies. 6. The Job Seeker opens a vacancy card. 7. The Job Seeker reviews title, company, responsibilities, requirements, salary, location, and employment conditions. 8. The Job Seeker decides that the vacancy is suitable for application.
Alternative Flow	1.1 Browse Recommended Vacancies 1. The Job Seeker is logged into hh.ru and opens the recommendation section. 2. The system displays recommended vacancies based on the profile and previous activity. 3. The Job Seeker opens a recommended vacancy card. 4. The flow returns to Normal Flow step 7.
Exceptions	1.0.E1 No Suitable Vacancies Found 1. The system returns no relevant vacancies. 2. The system prompts the Job Seeker to change keywords or filters. 3. The Job Seeker updates search parameters. 4. The flow returns to Normal Flow step 5. 1.0.E2 Vacancy Page Unavailable 1. The Job Seeker opens a vacancy that has been deleted or archived. 2. The system displays that the vacancy is unavailable. 3. The Job Seeker returns to search results.
Priority	High
Frequency of Use	Multiple times per day during active job search.
Other Information	Vacancy search and viewing can be performed without submitting an application. Login is required for personalized recommendations, favorites, and later application submission. The vacancy database is updated regularly, and vacancy descriptions contain enough information for the candidate to make a decision.
Business Rules	VAC-1, VAC-2, VAC-3, VAC-4, VAC-5, VAC-6, VAC-7

5. Use Case 2: Submit a Tailored Application

Field	Description
ID and Name	UC-2 Submit a Tailored Application
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	Employer
Description	After selecting a suitable vacancy in UC-1, the Job Seeker submits an application by choosing a resume, preparing or editing a cover letter, adding portfolio or project links if needed, and sending the application to the Employer through hh.ru.
Trigger	The Job Seeker clicks the Apply button on the selected vacancy page.
Preconditions	PRE-1. The selected vacancy is open for applications. PRE-2. The Job Seeker is logged into hh.ru. PRE-3. The Job Seeker has at least one resume in the account. PRE-4. Mandatory profile and resume fields are completed.
Postconditions	POST-1. The application record is created in hh.ru. POST-2. The Employer can see the application in the employer-side interface. POST-3. The application status is updated to sent or another equivalent system status. POST-4. The submitted application is visible to the Job Seeker in the responses section.
Normal Flow	2.0 Submit a Tailored Application 1. The Job Seeker clicks the Apply button on the selected vacancy page. 2. The system prompts the Job Seeker to choose a resume. 3. The Job Seeker selects the most relevant resume. 4. The system opens the application form and cover-letter field. 5. The Job Seeker writes a cover letter or edits the generated draft according to the vacancy requirements. 6. The Job Seeker manually adds portfolio or project links if they are important for the vacancy. 7. The Job Seeker reviews the application content. 8. The Job Seeker submits the application. 9. The system validates the required data. 10. The system sends the application to the Employer and displays a successful submission message.
Alternative Flow	2.1 Use Generated Cover Letter Draft 1. The Job Seeker chooses the built-in cover-letter generation option. 2. The system creates a draft letter. 3. The Job Seeker edits the draft to make it more specific to the vacancy. 4. The flow returns to Normal Flow step 6. 2.2 Add Portfolio Links Manually 1. The Job Seeker pastes GitHub, Behance, Figma, or other project links into the resume or cover-letter field. 2. The Job Seeker adds short explanations for selected projects. 3. The flow returns to Normal Flow step 7.
Exceptions	2.0.E1 Vacancy Closed Before Submission 1. The Employer closes or archives the vacancy before the final submission step. 2. The system informs the Job Seeker that the vacancy no longer accepts applications. 3. The application is not sent. 4. The Job Seeker returns to UC-1 Find a Vacancy. 2.0.E2 Incomplete Resume Data 1. The system detects missing mandatory fields in the selected resume. 2. The system prompts the Job Seeker to complete the missing data. 3. After correction, the flow returns to Normal Flow step 2.
Priority	High
Frequency of Use	Multiple times per day during active job search.
Other Information	The built-in letter generator is optional. Portfolio auto-import is treated as a potential improvement identified in Homework 1, so in the current use case portfolio links are added manually.
Assumptions	The selected vacancy is active, the Job Seeker has valid application materials, and the platform can deliver the application to the Employer.
Business Rules	VAC-1, VAC-6, APP-1, APP-2, APP-3, APP-4, APP-5, APP-6, APP-7

6. Use Case 3: Review Application Status

Field	Description
ID and Name	UC-3 Review Application Status
Created By	Team 9
Date Created	06/23/2026
Primary Actor	Job Seeker
Secondary Actors	Employer
Description	After submitting an application in UC-2, the Job Seeker checks the current status of the application and reviews any available employer update.
Trigger	The Job Seeker receives a notification about a status change, receives an employer message, or decides to check the application status manually.
Preconditions	PRE-1. A submitted application exists in the system. PRE-2. The application is visible to the Job Seeker in the responses section.
Postconditions	PRE-3. Status tracking is available. POST-1. The current application status is displayed. POST-2. Any available employer message or status update is displayed. POST-3. The application remains stored in the Job Seeker's responses section.
Normal Flow	3.0 Review Application Status 1. The Job Seeker receives a notification or opens hh.ru to check the application status. 2. The Job Seeker opens the responses section. 3. The system displays the list of submitted applications. 4. The Job Seeker selects the relevant application. 5. The system displays the current status and any available employer message. 6. The Job Seeker reviews the status and response information. 7. The Job Seeker decides on the next action outside the system.
Alternative Flow	3.1 Interview Invitation 1. The system shows that the application status is invitation. 2. The system displays interview details, contacts, or further instructions if the Employer provided them. 3. The Job Seeker reviews the information and prepares for the next hiring stage. 3.2 No Status Change Yet 1. The system shows that the application is still waiting for review. 2. The Job Seeker closes the response page and may continue searching for other vacancies. 3.3 Generic Rejection Comment 1. The system shows that the application status is rejected. 2. The system displays the employer's rejection comment. 3. The comment is generic and does not explain which skills or experience were missing. 4. The Job Seeker reviews the available comment and may revise the resume or continue searching.
Exceptions	3.0.E1 Message Cannot Be Loaded 1. The system fails to load the employer message or response details. 2. The system displays an error. 3. The Job Seeker refreshes the page or returns later.
Priority	High
Frequency of Use	Daily during active job search.
Other Information	A rejection usually includes an employer comment, but the comment can be generic and not actionable. More specific rejection reasons were identified as a possible improvement in Homework 1.
Assumptions	Employers update application statuses within a reasonable time, and hh.ru displays all response information that is available to the candidate.
Business Rules	APP-1, RESP-1, RESP-2, RESP-3, RESP-4, RESP-5, RESP-6

7. Conclusion

Business rules make the selected hh.ru use cases more precise because they define existing facts, restrictions on user actions, events that trigger system behavior, facts that can be inferred, and values that are calculated.

1. For **UC-1 Find a Vacancy**, business rules describe vacancy identity, employer ownership, search filters, vacancy availability, and relevance calculation.
2. For **UC-2 Submit a Tailored Application**, business rules define login requirements, application eligibility, mandatory resume data, generated cover-letter behavior, and application record creation.
3. For **UC-3 Review Application Status**, business rules define application statuses, status updates, rejection comments, generic rejection reasons, and interview invitation display.

The updated use cases now contain explicit references to business rule IDs. This improves traceability between user actions, system behavior, and business constraints, which helps business analysts, developers, and stakeholders maintain a shared understanding of the process.



Innopolis University

Homework 4: System Requirements for Use Cases

Business Analyst Role

Process: Searching and Applying for an Internship or Junior IT
Position Using hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Enrolment: June 30, 2026

1. Statement of Task

Objective: To write system requirements for the use cases from the previous homework using the study materials of Lecture 10 “Writing Excellent Requirements”.

The requirements are based on the use cases and business rules prepared in Homework 2 and Homework 3 for the **hh.ru** platform.

Key steps of the assignment:

1. Write system requirements for the previously defined use cases.
2. Explicitly specify backward traceability from each requirement to the related use case.
3. Use both system and user perspectives.
4. Give each requirement a clear unique identifier.
5. Write requirements in textual form using the keyword *shall*.
6. Keep the level of detail just enough for implementation and testing.
7. Group requirements where necessary and reuse repeated requirements.
8. Perform peer review of the requirements inside and outside the team.

2. Source Use Cases and Traceability Approach

The requirements are traced back to the three use cases from Homework 2:

- **UC-1 Find a Vacancy** - the Job Seeker searches for and selects a suitable vacancy.
- **UC-2 Submit a Tailored Application** - the Job Seeker applies to the selected vacancy.
- **UC-3 Review Application Status** - the Job Seeker checks the status of a submitted application.

Each requirement has a unique identifier in the format **SR-UCX-NN**, where:

- **SR** means System Requirement;
- **UCX** shows the related use case;
- **NN** is the requirement number inside the use case group.

The requirements also reference relevant business rules from Homework 3 where such connection is useful.

Use Case	Requirement IDs	Main Perspective
UC-1 Find a Vacancy	SR-UC1-01 – SR-UC1-06	User and system perspective
UC-2 Submit a Tailored Application	SR-UC2-01 – SR-UC2-08	User and system perspective
UC-3 Review Application Status	SR-UC3-01 – SR-UC3-06	User and system perspective

3. System Requirements

The table below contains 20 requirements. The wording follows the *shall* form, uses both user and system perspectives, and keeps each requirement focused on one observable capability or system response.

ID	Use Case	Perspective	Priority	Requirement Statement	Related BR
SR-UC1-01	UC-1	User	High	The Job Seeker shall be able to enter vacancy search keywords to search for relevant vacancies.	VAC-3
SR-UC1-02	UC-1	System	High	If the Job Seeker submits search keywords, the system shall display vacancies that match the active search parameters.	VAC-3, VAC-4
SR-UC1-03	UC-1	User	High	The Job Seeker shall be able to select search filters for city, specialization, experience level, employment type, salary, and work format.	VAC-3
SR-UC1-04	UC-1	System	High	If the Job Seeker changes search keywords or filters, the system shall update the list of matching vacancies.	VAC-5
SR-UC1-05	UC-1	System	High	If the Job Seeker opens a vacancy from the search results, the system shall display the vacancy title, employer, responsibilities, requirements, and employment conditions.	VAC-1, VAC-2
SR-UC1-06	UC-1	User	High	The Job Seeker shall be able to open the detailed view of any vacancy shown in the search results.	VAC-1, VAC-2
SR-UC2-01	UC-2	User	High	The Job Seeker shall be able to start an application from the page of a vacancy that is open for applications.	APP-3
SR-UC2-02	UC-2	System	High	If the Job Seeker is not logged into hh.ru, the system shall require authentication before displaying the application form.	APP-2
SR-UC2-03	UC-2	System	High	The system shall prevent application submission for a vacancy that is not open for applications.	VAC-6, APP-3
SR-UC2-04	UC-2	User	High	The Job Seeker shall be able to select one resume from the account for the selected application.	APP-1
SR-UC2-05	UC-2	System	High	If mandatory resume fields are missing, the system shall identify the missing fields and prompt the Job Seeker to complete them before application submission.	APP-4
SR-UC2-06	UC-2	User	High	The Job Seeker shall be able to edit the cover-letter text before final application submission.	APP-5, APP-6
SR-UC2-07	UC-2	System	High	If the Job Seeker submits a valid application, the system shall create an application record and send the application to the Employer.	APP-6, APP-7
SR-UC2-08	UC-2	System	High	If the application is submitted successfully, the system shall display a submission confirmation and show the application in the Job Seeker's responses section.	APP-7
SR-UC3-01	UC-3	User	High	The Job Seeker shall be able to open the responses section to view submitted applications.	RESP-1
SR-UC3-02	UC-3	System	High	The system shall display the list of submitted applications with their current statuses.	RESP-1, RESP-2
SR-UC3-03	UC-3	System	Medium	If an application is rejected and the Employer provided a comment, the system shall display the rejection comment for that application.	RESP-5
SR-UC3-04	UC-3	User	High	The Job Seeker shall be able to select a submitted application to view employer response details.	RESP-3
SR-UC3-05	UC-3	System	High	If the Employer changes the application status, the system shall update the status shown for that application in the Job Seeker's responses section.	RESP-3
SR-UC3-06	UC-3	System	High	If the Employer sends an interview invitation, the system shall display the interview details provided by the Employer for the selected application.	RESP-6

4. Reuse of Repeated Requirements

Some behavior is relevant to more than one use case. To avoid duplication, repeated requirements were written once and referenced through related use cases and business rules.

Requirement	Reusable Meaning	Where It Is Reused
SR-UC2-03	A vacancy must be open before an application can be submitted.	UC-1 vacancy selection and UC-2 application submission
SR-UC2-05	Missing mandatory resume data must be identified before submission.	UC-2 normal flow and UC-2 exception flow
SR-UC2-08	A successful submission must be confirmed and shown in the responses section.	UC-2 submission result and UC-3 later status review
SR-UC3-02	Submitted applications must be shown with current statuses.	UC-3 normal flow and UC-3 status-change alternative flows

5. Peer Review Summary

The requirements were reviewed by Team 9 members and one peer outside the team. The review focused on traceability, clarity, feasibility, verifiability, requirement granularity, and the use of the keyword *shall*. As a result of the review, several requirements were refined to avoid UI-specific wording, to make system responses more observable, and to cover missing submission outcomes.

Review Check	Result
Backward traceability	Each requirement is linked to UC-1, UC-2, or UC-3.
Shall form	All requirement statements use the keyword <i>shall</i> .
User and system perspectives	The set includes both “The Job Seeker shall” and “The system shall” statements.
Ambiguity check	Generic terms were replaced with consistent terms such as <i>Job Seeker</i> , <i>Employer</i> , <i>vacancy</i> , and <i>application</i> ; several statements were rewritten to describe observable results.
Verifiability check	Each requirement can be checked through UI behavior, system state, or application data.
Coverage check	Missing outcomes for successful application submission and rejection comment display were added to the requirement set.

6. Conclusion

The homework defines 20 system requirements for the three hh.ru use cases from the previous homework. The requirements are grouped by use case, uniquely labeled, written in *shall* form,



and traced back to the corresponding use cases.

The requirements cover the main candidate-side flow: searching for a vacancy, submitting a tailored application, and reviewing the application status. The set now also makes the submission result and rejection-comment handling explicit. The use of backward traceability and related business rule references helps connect user actions, system behavior, and business constraints in one consistent requirements set.



Innopolis University

Homework 5: Business Process Modelling

Business Analyst Role

New Feature: Portfolio Integration for hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Submission: July 7, 2026

1. Statement of Task

Objective: To model the business process that supports the new feature selected for the hh.ru platform using BPMN notation.

Homework 5 assignment: using the new feature of the selected application, draw the business process that demonstrates the main flow supporting this feature.

Selected application: hh.ru.

Selected new feature: Portfolio Integration into the Candidate Profile and Application Flow. This feature was selected because Homework 1 identified portfolio integration as the most promising improvement for students and junior candidates. Respondents explained that projects are the main proof of skills when they do not have commercial experience, while manual project links in the resume are unstructured and easy to overlook.

Process to model: A Job Seeker connects an external portfolio source, imports project information into hh.ru, selects relevant projects, saves a structured portfolio block, and attaches it to an application for a selected vacancy.

2. Business Process Scope

The modeled process focuses on the main business flow that enables a candidate to use the new portfolio integration feature during job application submission.

2.1 Business Goal

The business goal of the process is to make a candidate's practical projects visible to the Employer in a structured and reusable form. This is especially important for students and Junior IT candidates, because they often have fewer commercial work experience entries and need another reliable way to prove their skills.

2.2 Participants, Pools, and Lanes

The corrected BPMN model separates independent participants into different pools. The hh.ru pool is additionally divided into lanes, because the user interface and backend services perform different types of tasks.

Pool / Lane		BPMN role	Responsibility in the process
Job Seeker		External participant / pool	Initiates the portfolio integration, selects a portfolio source, grants access, reviews imported projects, adds manual links if needed, and applies to a vacancy.
hh.ru / Candidate Profile UI		Lane inside hh.ru pool	Displays supported sources, imported projects, manual entry form, and the saved portfolio block.
hh.ru / Integration and Application Services		Lane inside hh.ru pool	Requests authorization, imports portfolio data, validates project items, saves the portfolio block, creates the application record, and sends the application to the Employer.
External Platform	Portfolio	External participant / pool	Authorizes access and returns project data from services such as GitHub, Behance, Figma, or similar platforms.
Employer		External participant / pool	Receives the submitted application and reviews the resume together with the attached portfolio block.

2.3 Trigger, Preconditions, and Result

Element	Description
Trigger	The Job Seeker decides to improve the application by attaching structured portfolio projects.
Preconditions	The Job Seeker is logged into hh.ru; the Job Seeker has a resume; the selected vacancy is open for applications; at least one external portfolio source can be connected or manual project links can be added.
Postconditions	A structured portfolio block is saved in the candidate profile; the selected application is submitted with a portfolio snapshot; the Employer can review the attached portfolio together with the resume and cover letter.



3. BPMN Main Flow Diagram

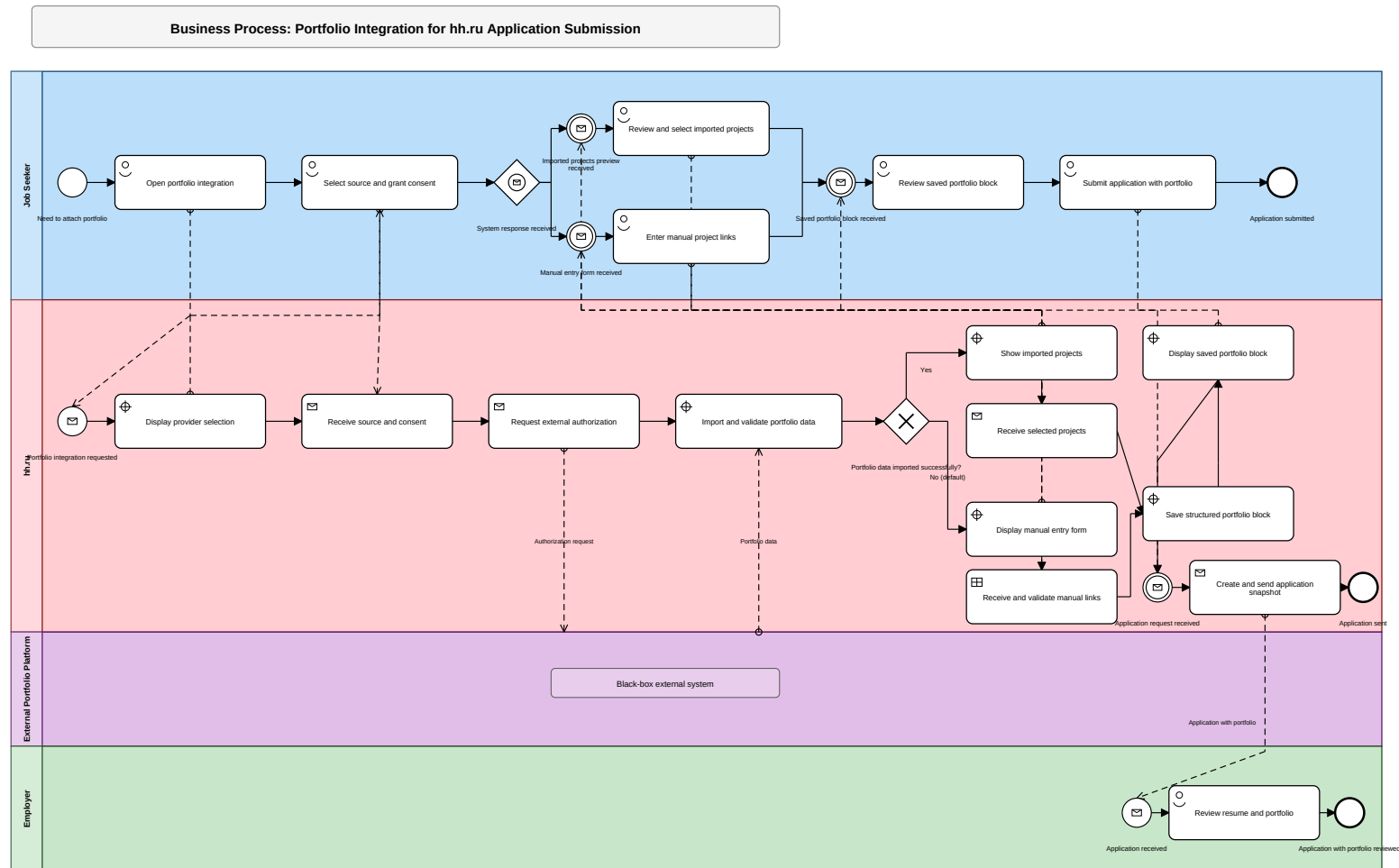


Figure 1: BPMN diagram for the Portfolio Integration process

Diagram note: Solid arrows show sequence flows inside one pool. Dashed arrows show message flows between independent participants. The model separates the Job Seeker, hh.ru, the External Portfolio Platform, and the Employer into different pools, while hh.ru is divided into Candidate Profile UI and Integration and Application Services lanes. The import failure branch goes through manual link entry and validation before saving the portfolio block.

4. Main Flow Description

The BPMN diagram demonstrates the main business process supporting the new portfolio integration feature.

1. The Job Seeker opens the portfolio integration function in hh.ru.
2. hh.ru displays supported external portfolio sources, such as GitHub, Behance, Figma, or similar services.
3. The Job Seeker selects a portfolio source and grants access to project data.
4. hh.ru receives the selected source and consent, then requests authorization from the external portfolio platform.
5. The external portfolio platform authorizes access and returns available project data to hh.ru.
6. hh.ru imports the portfolio data and validates the imported items according to business rules.
7. If the imported data is valid, hh.ru shows imported projects to the Job Seeker.
8. The Job Seeker reviews the imported projects, selects the most relevant ones, and edits descriptions for the target vacancy.
9. If the import fails or the imported data is invalid, hh.ru displays a manual entry form and the Job Seeker enters project links manually.
10. hh.ru validates either the selected imported projects or the manual links, then saves the result as a structured portfolio block in the candidate profile.
11. hh.ru displays the saved portfolio block to the Job Seeker.
12. The Job Seeker reviews the saved portfolio block and submits the application with the attached portfolio.
13. hh.ru receives the application request, creates an application record with a portfolio snapshot, and sends the application to the Employer.
14. The Employer reviews the resume together with the attached portfolio block.

5. Alternative and Exception Flow

The diagram includes one controlled alternative branch. If imported portfolio data is invalid, inaccessible, empty, unsupported, or cannot be imported, the process does not stop. Instead, hh.ru displays a manual entry form, the Job Seeker enters project links manually, and hh.ru validates these links before saving the same structured portfolio block.

Situation	System behavior
Authorization denied	hh.ru explains that access was not granted and lets the Job Seeker choose another source or continue with manual project links.
No public projects found	hh.ru informs the Job Seeker that no accessible projects were imported and displays the manual entry form.
Duplicated or invalid links	hh.ru filters duplicates, marks invalid links, validates manual input, and asks the Job Seeker to review the portfolio block before using it.
External platform unavailable	hh.ru shows an import error and allows the Job Seeker to continue with manual project links, so the vacancy application is not blocked.

6. BPMN Elements Used

BPMN element	How it is used in the process
Pools	Separate independent participants: Job Seeker, hh.ru, External Portfolio Platform, and Employer.
Lanes	The hh.ru pool is divided into Candidate Profile UI and Integration and Application Services to show ownership of tasks inside the platform.
Start Events	The process starts when the Job Seeker decides to attach portfolio projects and hh.ru receives the integration request.
User Tasks	The Job Seeker selects a source, grants access, reviews imported projects or enters manual links, reviews the saved portfolio block, and submits the application. The Employer reviews the received application.
Service Tasks	hh.ru displays UI screens, imports portfolio data, saves the structured portfolio block, and creates the application record.
Receive and Send Tasks	hh.ru receives source, consent, selected projects, manual links, and application request; hh.ru sends authorization requests and the final application to the Employer.
Business Rule Tasks	hh.ru validates imported project items and manually entered links according to rules for accessibility, duplicates, supported sources, and required project metadata.
Exclusive Gateways	The process checks whether imported portfolio data is available and valid. Only one branch is selected: imported projects or manual fallback. The Job Seeker also follows the corresponding preview or manual entry path shown by hh.ru.
Sequence Flows	Solid arrows show the order of actions inside the same pool.
Message Flows	Dashed arrows show interactions between different participants, such as hh.ru requesting external authorization or sending the application to the Employer.
Data Objects	Portfolio data, the structured portfolio block, and the final application snapshot show the key information created and used by the process.
End Events	The process ends when the application has been submitted and the Employer has reviewed the application with the attached portfolio.

7. Connection with Previous Homeworks

Previous work	Connection to Homework 5
Homework 1	Interview results confirmed that students without commercial experience need a structured way to present projects. Portfolio integration was selected as the most promising improvement.
Homework 2	The process extends UC-2 Submit a Tailored Application, where the Job Seeker submits an application with resume, cover letter, and portfolio materials.
Homework 3	Business rules are used in the validation steps: hh.ru checks portfolio availability, completeness, duplicates, source support, and suitability before saving the block.
Homework 4	The process supports system requirements related to application submission, application record creation, and presenting application materials to the Employer.

8. Conclusion

Homework 5 models the main BPMN business process for the new hh.ru portfolio integration feature. The corrected process separates independent participants into different pools and uses message flows for interactions between them. The hh.ru pool is divided into UI and backend service lanes, which makes task ownership clearer.

The model supports the main user value identified in previous homeworks: students and junior candidates can present practical projects in a structured way instead of manually pasting project links into a resume or cover letter. The BPMN model also includes a manual fallback branch, so the application process remains usable even when external portfolio import fails.



Innopolis University

Homework 6: Throwaway Prototype and Dialog Map

Business Analyst Role

Application: hh.ru

Feature: Portfolio Integration into Candidate Profile and Application Flow

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Submission: July 15, 2026

1. Statement of Task

Objective: To create an electronic throwaway mock-up for the selected application feature and to draw a dialog map that defines transitions between screens.

Homework 6 assignment:

1. Use previous homeworks as input: hypotheses, use cases, business rules, functional requirements, and business processes.
2. Draw a dialog map to define transitions between screens.
3. Draw an electronic throwaway mock-up.
4. Embed the mock-up into the existing system interface of the selected application.
5. Keep previous requirements for homework properties.

Selected application: hh.ru.

Selected feature for the prototype: Portfolio Integration into the Candidate Profile and Application Flow.

Prototype type: horizontal electronic throwaway mock-up. The prototype is intended to clarify requirements and navigation, not to become production code.

2. Input Summary and Traceability Basis

Homework 6 uses the results of Homeworks 1–5 as design input. The full source materials are documented in the combined Team 9 file `main.tex`; this section summarizes only the inputs that directly shaped the dialog map and the electronic mock-up.

Selected feature for the prototype: Portfolio Integration into the Candidate Profile and Application Flow.

Source	Input used in Homework 6
Homework 1	Hypothesis 2 (Portfolio Integration) was confirmed. Students without commercial experience need a structured way to present GitHub, Behance, Figma, and similar project evidence inside hh.ru instead of unstructured manual links.
Homework 2 and 3	UC-2 Submit a Tailored Application defines the base application flow: resume selection, cover letter, application review, and submission. Related business rules APP-1–APP-7 constrain authentication, vacancy availability, mandatory resume data, and successful application delivery.
Homework 4	System requirements SR-UC2-01–SR-UC2-08 support the submission part of the prototype, especially starting an application, validating mandatory data, creating an application record, and showing the result in the responses section.
Homework 5	The BPMN process for portfolio integration defines the new screens and transitions: source selection, external import, project review, manual fallback, saved portfolio block, and application submission with portfolio snapshot.

How this input is reflected in Homework 6:

- The dialog map extends UC-2 with portfolio-specific states S2–S10 while keeping the existing application start and submission ending.
- The mock-up embeds the new portfolio block into the familiar hh.ru response form instead of designing a separate standalone application.
- Screen 4 covers the Homework 5 alternative branch for import failure or invalid data.
- Screens 5 and 6 connect to APP-6, SR-UC2-07, and SR-UC2-08 and then to UC-3 through the responses section.

3. Prototype Scope and Questions

The prototype focuses on the uncertain part of the future feature: how the Job Seeker connects a portfolio source, chooses relevant projects, and attaches a structured portfolio block to the application without leaving the familiar hh.ru application flow.

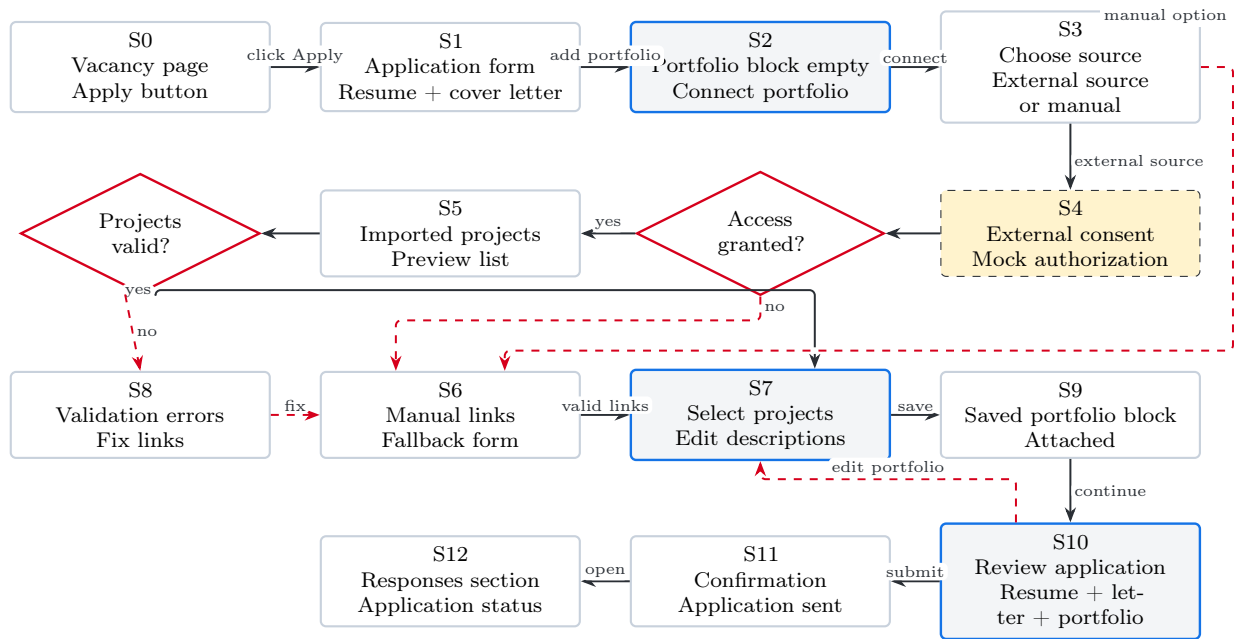
Questions that the prototype should answer:

1. Does the Job Seeker understand where to add a portfolio inside hh.ru?
2. Is the transition from resume selection to portfolio attachment logical?
3. Are external-source import and manual link entry both visible enough?
4. Does the user understand that selected projects will be sent with the application?
5. Are there dead ends if authorization fails or no projects are imported?

What is intentionally faked in the mock-up:

- external authorization with GitHub, Behance, Figma, or another platform;
- imported project data and repository statistics;
- validation of project links;
- application delivery to the Employer;
- final visual design details such as exact spacing, fonts, and production colors.

4. Dialog Map



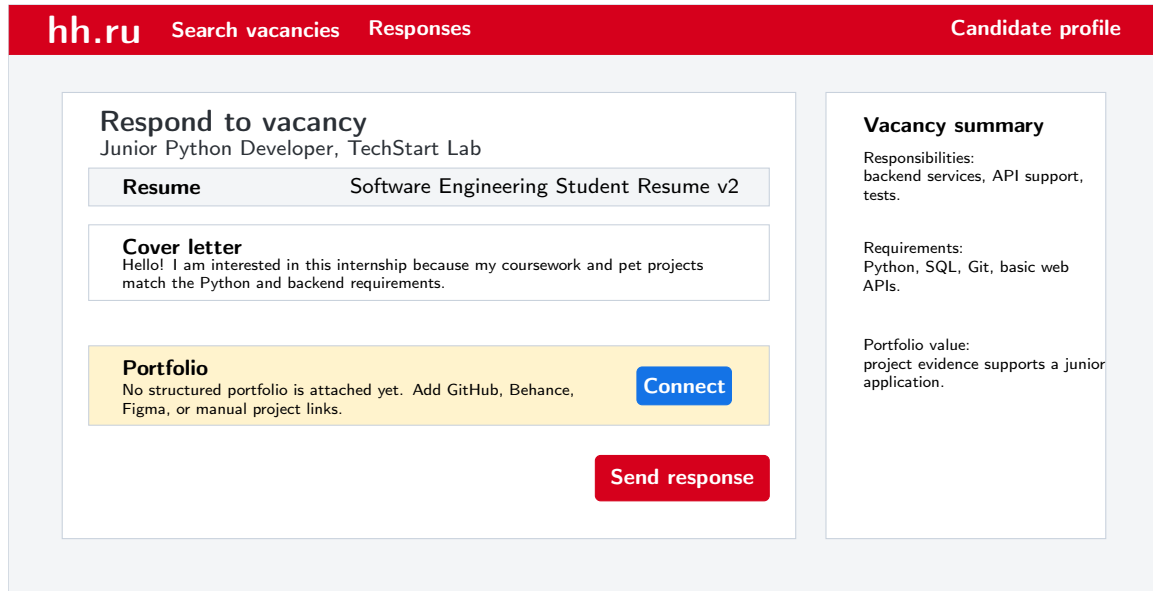
Dialog map note. The map represents screen transitions, not backend sequence logic. BPMN from Homework 5 was used as a process input, while this diagram focuses on user navigation and UI states.

Figure 2: Dialog map for portfolio integration in hh.ru

5. Electronic Throwaway Mock-up

The following wireframes embed the new portfolio integration feature into a simplified hh.ru-like interface. The screens intentionally look like low-fidelity electronic mock-ups: they show layout, controls, data blocks, and navigation, while avoiding unnecessary visual polish.

5.1 Screen 1: Application Form with Portfolio Entry Point



hh.ru

Search vacancies

Responses

Candidate profile

Respond to vacancy

Junior Python Developer, TechStart Lab

Resume

Software Engineering Student Resume v2

Cover letter

Hello! I am interested in this internship because my coursework and pet projects match the Python and backend requirements.

Portfolio

No structured portfolio is attached yet. Add GitHub, Behance, Figma, or manual project links.

Connect

Send response

Vacancy summary

Responsibilities:

backend services, API support, tests.

Requirements:

Python, SQL, Git, basic web APIs.

Portfolio value:

project evidence supports a junior application.

Figure 3: Mock-up screen 1: portfolio entry point inside the hh.ru application form

5.2 Screen 2: Select Portfolio Source

hh.ru
Search vacancies
Responses
Candidate profile

Connect portfolio source

Choose where hh.ru should import public project information from. External authorization is simulated in this throwaway mock-up.

GitHub
Repositories and
README descriptions

Behance
Design cases and
visual projects

Figma
Prototype and
interface links

Manual links
Use this option if import fails or the source is not supported.

Cancel
Continue

Only public project title, description, technologies, and link are imported. Private repositories and personal data are not imported.

Figure 4: Mock-up screen 2: portfolio source selection embedded in hh.ru

5.3 Screen 3: Review Imported Projects

hh.ru

Search vacancies

Responses

Candidate profile

Imported projects from GitHub

Select projects relevant to Junior Python Developer. The employer will see only selected projects.

☐

Job search analytics dashboard

Technologies: Python; SQL; pandas

selected

☐

REST API task tracker

Technologies: FastAPI; PostgreSQL; tests

selected

☐

University course website

Technologies: HTML; CSS; JavaScript

not selected

Short note for employer

These projects demonstrate backend basics, SQL, tests, and data analysis.

Back

Save

Figure 5: Mock-up screen 3: imported project review and selection

5.4 Screen 4: Manual Fallback for Project Links

hh.ru

Search vacancies

Responses

Candidate profile

Add portfolio manually

Use this fallback when authorization is denied, no public projects are found, or the external platform is unavailable.

Import from GitHub failed in this prototype. You can still attach links manually.

Project	Project link	Short description
Project	Project link	Short description
Project	Project link	Short description

+ Add project

Back

Validate and save

Figure 6: Mock-up screen 4: manual fallback path with validation before saving

5.5 Screen 5: Portfolio Attached to Application

hh.ru

Search vacancies

Responses

Candidate profile

Review before sending

Junior Python Developer - TechStart Lab

Resume: Software Engineering Student Resume v2

Cover letter

Editable text connected to vacancy requirements.

Portfolio block attached

2 selected projects

Evidence: Python, SQL, FastAPI, tests

Edit

Send response

Employer view

Resume

Cover letter

Structured portfolio

Project title, link, short description, technologies

Ready to submit

Figure 7: Mock-up screen 5: selected projects attached to the hh.ru response

5.6 Screen 6: Submission Confirmation

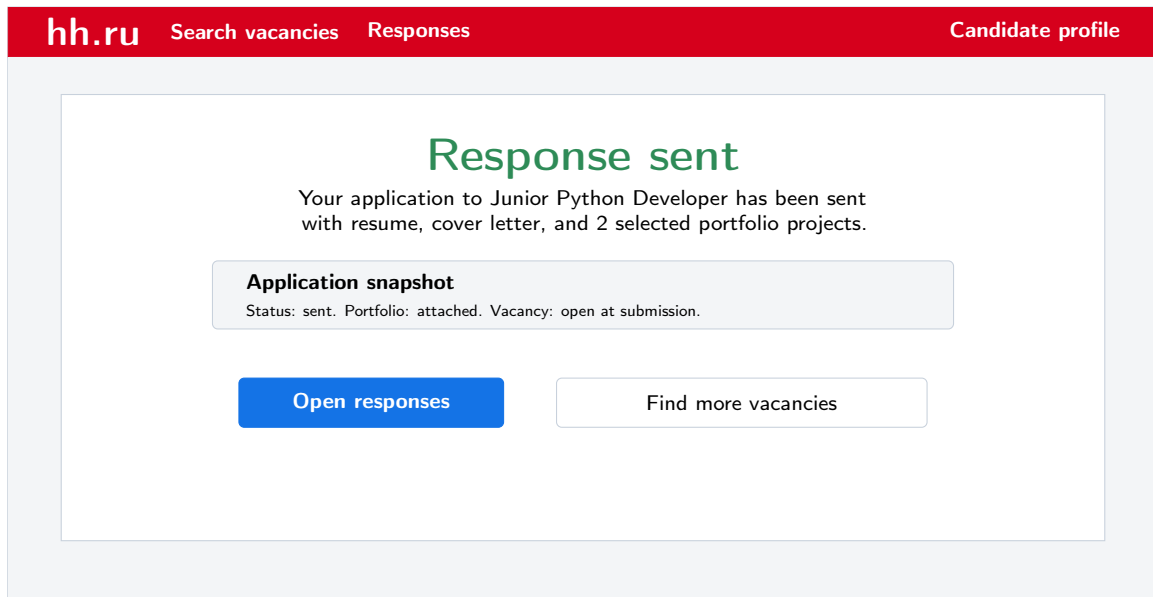


Figure 8: Mock-up screen 6: confirmation and transition to the responses section

6. Dialog Map to Mock-up Traceability

Dialog state	map	Mock-up screen	Requirement or process connection
S0–S2		Screen 1	Extends UC-2: the user starts an application and sees a portfolio entry point inside the application form.
S3–S4		Screen 2	Supports the Homework 5 process step where the Job Seeker selects an external portfolio source and grants consent.
S5–S7		Screen 3	Supports project import, review, selection, and short description editing before saving a structured block.
S6–S8		Screen 4	Covers the alternative branch from Homework 5: import failure or invalid data leads to manual link entry.
S9–S10		Screen 5	Connects to APP-6, SR-UC2-07, and SR-UC2-08: the valid application is reviewed and can be sent.
S11–S12		Screen 6	Connects UC-2 to UC-3: after submission, the application can be opened in the responses section.

7. Prototype Evaluation Script

The prototype should be evaluated with users from the same target group: undergraduate students applying for internships or Junior IT positions.

7.1 Evaluation Tasks

1. Start from the vacancy application form and attach a portfolio from GitHub.
2. Select two projects that are relevant to the vacancy.
3. Edit the short portfolio note before saving.
4. Return to the application review screen and check what the Employer will see.
5. Submit the response and open the responses section.
6. Repeat the flow using manual links when import fails.

7.2 Evaluation Questions

1. Was it clear where to add a portfolio in the application flow?
2. Did the source selection screen provide enough options?

3. Were you ever unsure what to do next?
4. Does the imported project preview contain enough information for choosing projects?
5. Is the manual fallback visible and understandable?
6. Would this portfolio block make your application stronger than plain links in a resume?
7. What functions are missing from this prototype?

8. Risks and Design Decisions

Risk	Mitigation in the prototype
Users may think the prototype is final UI design.	The mock-up is intentionally low-fidelity and described as throwaway.
Users may focus on colors and fonts instead of functionality.	Evaluation questions focus on navigation, task completion, missing states, and clarity.
External import may fail in the real system.	The dialog map and mock-up include a manual fallback path.
Employers may need a concise portfolio view.	The mock-up shows a compact application snapshot with only selected projects.
The flow may become too long for quick applications.	The dialog map keeps portfolio setup reusable: once saved, the block can be attached again later.

9. Conclusion

Homework 6 presents a dialog map and an electronic throwaway mock-up for the portfolio integration feature in hh.ru. The prototype is based on the confirmed portfolio integration hypothesis, UC-2 Submit a Tailored Application, related business rules, system requirements, and the BPMN process from Homework 5.

The main value of the prototype is that it makes the future feature easier to discuss with users and stakeholders. It shows where the portfolio block appears in the existing application flow, how external import and manual fallback work, and how selected projects become part of the submitted application snapshot.



Innopolis University

Homework 7: System Requirements Prioritization

Business Analyst Role

Process: Searching and Applying for an Internship or Junior IT
Position Using hh.ru

Team Number: Team 9

Team Members:

- Student 1: Bashirov Niyaz
- Student 2: Aleksandrova Marina
- Student 3: Iarovenko Arina
- Student 4: Latipova Aliya
- Student 5: Kolesnikov Arsenii
- Student 6: Belyaev Vladimir

Date of Enrolment: July 17, 2026

1. Statement of Task

Objective: To prioritize functional system requirements from Homework 4 using the Three-Level Scale technique from Lecture 15.

Key steps of the assignment:

1. Select exactly 10 functional requirements from the 20 requirements defined in Homework 4.
2. Group the selected requirements logically by their related use cases.
3. Consider direct dependencies between the selected requirements.
4. Assign each selected requirement one of three priority levels: High, Medium, or Low.
5. Create the table “System Requirements Prioritization” with the columns “Req. label” and “Priority”.

The source requirements are traced to the following use cases:

- UC-1 Find a Vacancy;
- UC-2 Submit a Tailored Application;
- UC-3 Review Application Status.

2. Selected System Requirements

From the 20 functional requirements prepared in Homework 4, the following **10 requirements** were selected. They cover the complete candidate-side process: finding a vacancy, submitting an application, and reviewing the application result.

Table 11: Selected System Requirements

Req. label	Requirement name	Requirement statement
Group 1: UC-1 Find a Vacancy		
SR-UC1-01	Enter Search Keywords	The Job Seeker shall be able to enter vacancy search keywords to search for relevant vacancies.
SR-UC1-02	Display Matching Vacancies	If the Job Seeker submits search keywords, the system shall display vacancies that match the active search parameters.
SR-UC1-06	Open Vacancy Detailed View	The Job Seeker shall be able to open the detailed view of any vacancy shown in the search results.
SR-UC1-05	Display Vacancy Details	If the Job Seeker opens a vacancy from the search results, the system shall display the vacancy title, employer, responsibilities, requirements, and employment conditions.
Group 2: UC-2 Submit a Tailored Application		
SR-UC2-01	Start Application	The Job Seeker shall be able to start an application from the page of a vacancy that is open for applications.
SR-UC2-02	Require Authentication	If the Job Seeker is not logged into hh.ru, the system shall require authentication before displaying the application form.
SR-UC2-04	Select Resume	The Job Seeker shall be able to select one resume from the account for the selected application.
SR-UC2-07	Create and Send Application	If the Job Seeker submits a valid application, the system shall create an application record and send the application to the Employer.
Group 3: UC-3 Review Application Status		
SR-UC3-02	Display Applications and Statuses	The system shall display the list of submitted applications with their current statuses.
SR-UC3-03	Display Rejection Comment	If an application is rejected and the Employer provided a comment, the system shall display the rejection comment for that application.



3. System Requirements Dependencies

Dependencies were analyzed before assigning priorities. In Table 12, a “+” means that the requirement in the row directly depends on the requirement in the column. “N/A” marks the same requirement on both axes. To keep the matrix readable, only the strongest direct prerequisite is shown for each requirement.

System Requirements Dependencies

System require- ment	SR-UC1-01	SR-UC1-02	SR-UC1-06	SR-UC1-05	SR-UC2-01	SR-UC2-02	SR-UC2-04	SR-UC2-07	SR-UC3-02	SR-UC3-03
SR-UC1-01	N/A									
SR-UC1-02	+	N/A								
SR-UC1-06		+	N/A							
SR-UC1-05			+	N/A						
SR-UC2-01				+	N/A					
SR-UC2-02					+	N/A				
SR-UC2-04						+	N/A			
SR-UC2-07							+	N/A		
SR-UC3-02								+	N/A	
SR-UC3-03									+	N/A

Table 12: Dependency matrix for the 10 selected requirements

4. Prioritization Approach

The Three-Level Scale technique was applied using importance, urgency, business value, and dependencies. Dependencies were used to determine implementation order, but a predecessor was not automatically classified as High priority.

- **High priority** – required to complete the main search or application-submission flow, or required as a predecessor for another essential function.
- **Medium priority** – important supporting functionality that improves the flow or connects one major stage with another, but can be delivered after the first essential requirements.
- **Low priority** – useful for a limited scenario and can be postponed without blocking vacancy search, application submission, or basic status review.

High Medium Low

5. System Requirements Prioritization

Req. label	Priority
Group 1: UC-1 Find a Vacancy	
SR-UC1-01	High
SR-UC1-02	High
SR-UC1-06	Medium
SR-UC1-05	Medium
Group 2: UC-2 Submit a Tailored Application	
SR-UC2-01	High
SR-UC2-02	High
SR-UC2-04	Medium
SR-UC2-07	High
Group 3: UC-3 Review Application Status	
SR-UC3-02	High
SR-UC3-03	Low

5.1 Priority Distribution

- High – 6 requirements;
- Medium – 3 requirements;
- Low – 1 requirement.

5.2 Priority Rationale

High-priority requirements. SR-UC1-01 and SR-UC1-02 provide the minimum vacancy-search capability. SR-UC2-01, SR-UC2-02, and SR-UC2-07 are required to start, authorize, create, and send an application. SR-UC3-02 is the core requirement of UC-3 because, without a list of submitted applications and their current statuses, the Job Seeker cannot review application progress.

Medium-priority requirements. SR-UC1-06 allows the Job Seeker to open a vacancy detailed view, while SR-UC1-05 displays the information needed for an informed application decision. Both requirements are important for a complete vacancy-review experience, but the basic vacancy-search results can be delivered first. SR-UC2-04 allows the Job Seeker to choose a resume; in an early release, the system could use the only available or a previously selected resume.

Low-priority requirement. SR-UC3-03 applies only to rejected applications for which the Employer has provided a comment. It does not block vacancy search, application submission, or basic status review, so it can be postponed to a later iteration.

Dependency note. Dependencies determine the safe implementation order, but they do not automatically make every predecessor High priority. A Medium-priority prerequisite may still be implemented before a High-priority dependent requirement when the release sequence requires it.

6. Recommended Implementation Sequence

The dependency analysis produces the following logical order:

SR-UC1-01 → SR-UC1-02 → SR-UC1-06 → SR-UC1-05 → SR-UC2-01 →
SR-UC2-02 → SR-UC2-04 → SR-UC2-07 → SR-UC3-02 → SR-UC3-03.

The sequence also follows the use-case dependency chain: UC-1 must provide a suitable vacancy before UC-2 can submit an application, and UC-2 must create the application before UC-3 can display its status or rejection comment.

7. Conclusion

Exactly 10 functional requirements were selected from Homework 4 and grouped by UC-1, UC-2, and UC-3. Their direct dependencies were documented in a dependency matrix, and each requirement was assigned High, Medium, or Low priority using the Three-Level Scale technique.

The final distribution contains six High-priority requirements, three Medium-priority requirements, and one Low-priority requirement. High priority is reserved for the minimum search, application-submission, and status-review capabilities. Opening and displaying vacancy details and selecting a resume are Medium priority, while the rejection-comment scenario is Low priority. This distribution differentiates the requirements clearly while preserving a complete, dependency-aware implementation sequence.